VIT HACKATHON | 24-HOUR CHALLENGE

Duplicate Defect Finder & Bug Report Enhancer

AI-Powered Bug Triage & Quality Enhancement System

PRESENTED BY

Neural Ninjas

Duplicate Bugs Are Killing Productivity

Wasted Engineering Time

Engineers waste ~30% of their time triaging and closing duplicate reports instead of coding.

Poor Report Quality

Vague descriptions and missing reproduction steps delay critical bug fixes by days.

Issue Tracker Chaos

No intelligent triage results in hundreds of overlapping tickets, making prioritization impossible.

40–60%

of daily bug reports are duplicates

02

One Tool. Three Superpowers.

A unified engine that transforms how development teams handle defect tracking.

Detect Duplicates

Instantly identifies duplicate reports with a semantic confidence score, preventing redundant work before it starts.

Technique: Vector Similarity

Auto Cluster

Groups related issues into logical clusters (e.g., "Login", "UI"), revealing hidden patterns in the chaos.

Technique: DBSCAN Clustering

Enhance Quality

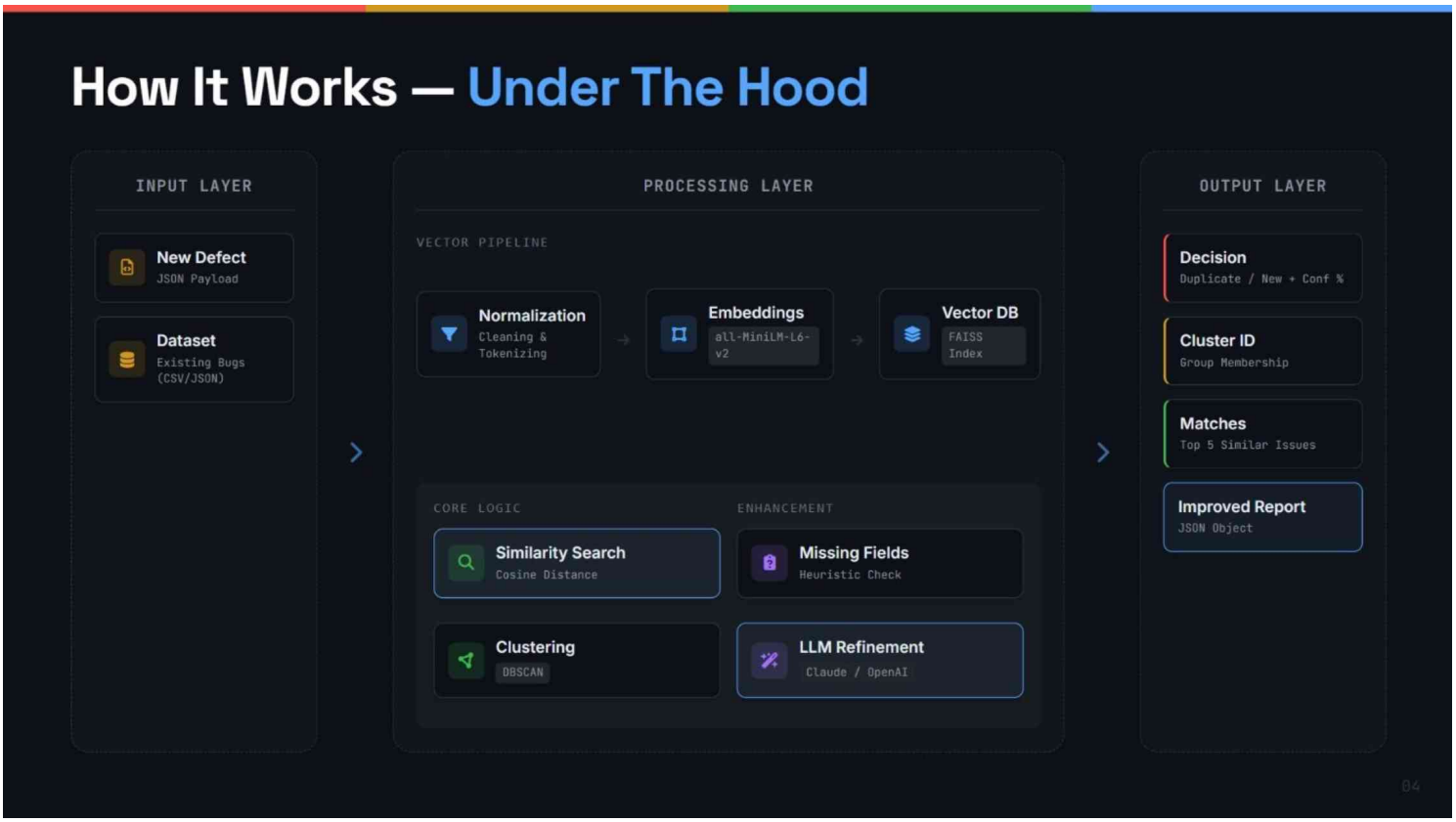
Uses LLMs to rewrite vague reports, filling in missing steps and suggesting severity levels automatically.

Technique: GenAI Refinement

SYSTEM WORKFLOW

New Bug Report → AI Engine Core → Decision + Enhanced Report

03


How It Works — Under The Hood
INPUT LAYER
New Defect
JSON Payload
Dataset
Existing Bugs (CSV/JSON)
PROCESSING LAYER
VECTOR PIPELINE
Normalization
Cleaning & Tokenizing
Embeddings
all-MiniLM-L6-v2
Vector DB
FAISS Index
CORE LOGIC
Similarity Search
Cosine Distance
Clustering
DBSCAN
ENHANCEMENT
Missing Fields
Heuristic Check
LLM Refinement
Claude / OpenAI
OUTPUT LAYER
Decision
Duplicate / New + Conf %
Cluster ID
Group Membership
Matches
Top 5 Similar Issues
Improved Report
JSON Object
04

Clean Inputs. Actionable Outputs.

RAW INPUT

SPARSE DATA

```
{
"defect_id": "NEW-001",
"title": "app crash",
"description": "it crashes",
"steps": null,
"expected": null,
"actual": null,
"environment": null,
"logs": null

}
```

→

AI RESPONSE

ENRICHED DATA

```
{
"decision": "duplicate",
"confidence": 0.92,
"top_matches": [
  {"id": "BUG-3921", "score": 0.92},
  {"id": "BUG-3805", "score": 0.88}
],
"cluster_id": "C1",
"improved_report": {
  "title": "Application crashes on dashboard load",
  "summary": "...",
  "missing_fields": ["steps", "environment"],
  "suggested_severity": "High"

}
}
```

05

Smart Duplicate Detection with Thresholds

Semantic vector search ensures high precision matching.

≥ 85% — **DUPLICATE**
Auto-linked as duplicate. Confidence is high enough to skip manual triage.

50–85% — **POSSIBLE DUPLICATE**
Flagged for review. Engineer sees "Did you mean this?" suggestion.

< 50% — **NEW DEFECT**
Unique issue. Passed to classification engine for processing.

Safety Guard: Never auto-resolves below 95% threshold.

Powered by sentence-transformers + FAISS

Top Matches Preview

Input: "App crashing on dashboard load" QUERY VECTOR

SEARCH RESULTS

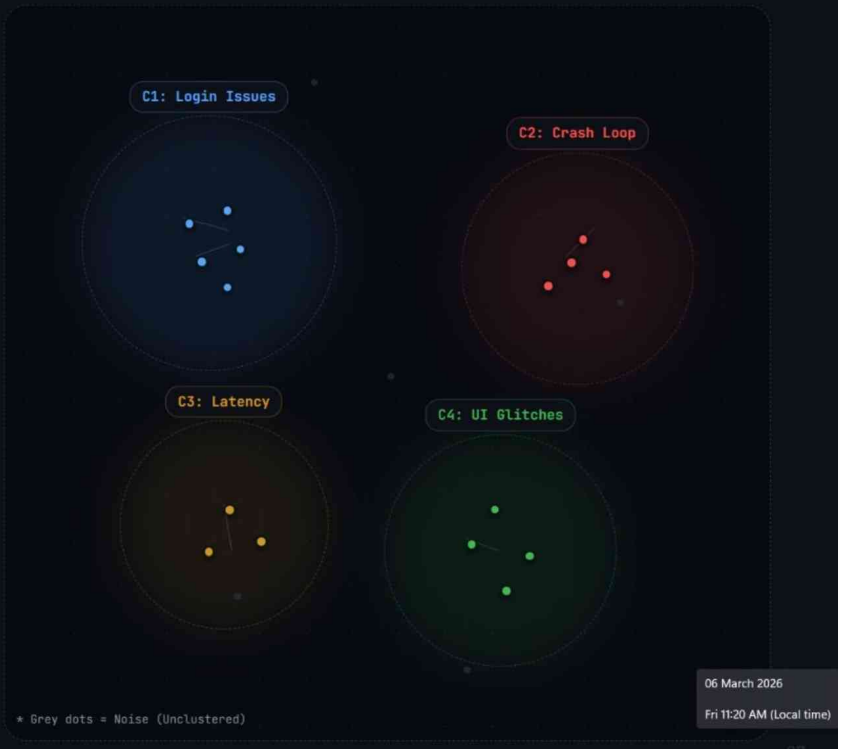
BUG-3921	Dashboard crashes when loading user data widgets	92%
BUG-3805	Application freeze on startup dashboard	88%
BUG-3112	Admin panel performance degradation	65%
BUG-2998	User profile image fails to load	32%
BUG-2840		

Auto-Group Related Bugs with DBSCAN

Spot patterns across hundreds of reports instantly by grouping semantic vectors.

- **Cluster C1: Login/Auth**
  5 Linked Reports
- **Cluster C2: Dashboard Crashes**
  4 Linked Reports
- **Cluster C3: Performance**
  3 Linked Reports
- **Cluster C4: UI/Display**
  4 Linked Reports

Tech: DBSCAN + Sentence Embeddings



07

Transform Vague Reports into Actionable Ones

Generative AI fills gaps, structures data, and assigns metadata automatically.

Raw Submission POOR QUALITY

TITLE

app doesnt work

DESCRIPTION

it crashed again when i tried to open it

MISSING CRITICAL FIELDS

- × Steps to Reproduce
- × Expected Result
- × Environment / Device

LLM Processing

AI Enhanced READY FOR DEV

REFINED TITLE

Application crashes on startup after update v2.1

STRUCTURED SUMMARY

App consistently crashes on launch following the v2.1 update across Android devices based on context.

INFERRED STEPS

1. Install v2.1 update
2. Launch application
3. Observe immediate crash

FLAGGED MISSING

Logs, Environment Details

AUTO-SEVERITY

High

08

Built With Battle-Tested Technology

A modern, scalable architecture designed for speed and accuracy.

Backend & AI

- **Python + FastAPI**
  High-performance async API
- **sentence-transformers**
  SBERT Embedding Generation
- **FAISS Vector DB**
  Similarity Search Engine
- **DBSCAN**
  Density-Based Clustering
- **Claude API**
  LLM Text Enhancement

Frontend

- **React.js**
  Component-based UI
- **Tailwind CSS**
  Utility-first Styling
- **Recharts**
  Data Visualization
- **lucide-react**
  Clean Iconography

Data Sources

- **Bugzilla**
  Mozilla Bug Reports
- **GitBugs Dataset**
  GitHub Issue Archive
- **Jira Issue Reports**
  Apache Project Issues

100k+ Training Samples

Real-World Data. Real-World Results.

Trained on diverse datasets to ensure robust generalization across domains.

Bugzilla

Mozilla's public bug tracking database containing years of browser defects.

150k Reports

GitBugs

Curated collection of issue reports from top open-source GitHub repositories.

85k Issues

Jira / Apache

Complex enterprise-grade issue tickets from the Apache Software Foundation.

45k Tickets

DATA PROCESSING PIPELINE

Normalization
Cleaning HTML tags, stack traces & special chars

Field Mapping
Aligning disparate schemas to unified JSON

Field Detection
identifying missing steps & environment data

Embedding
Generating 384-d vectors via MiniLM-L6

280k+ Processed / 42k Duplicates Found

Simple REST API. Powerful Results.

Seamless integration with your existing issue tracking workflow.

POST `/api/v1/check-duplicate`

Analyzes a new defect report against the vector database to find semantic duplicates.

Response Example

```
"decision": "duplicate",
"confidence": 0.92,
"top_matches": [
  { "id": "BUG-104", "score": 0.92 }
],
"cluster_id": "C1"
```

POST `/api/v1/enhance-report`

Uses LLM to generate improved titles, summaries, and identify missing fields.

Response Example

```
"original_score": 0.45,
"improved_report": {
  "title": "App crash on login...",
  "missing_fields": ["steps", "env"]
},
"suggested_severity": "High"
```

GET `/api/v1/reports`

Retrieves list of processed defects with their enrichment status and metadata.

Response Example

```
[
  {
    "id": "NEW-001",
    "status": "processed",
    "is_duplicate": true
  },
  ...
]
```

GET `/api/v1/clusters`

Returns active defect clusters with aggregated insights and member counts.

Response Example

```
"clusters": [
  {
    "id": "C1",
    "topic": "Login Failure",
    "count": 5,
    "members": ["BUG-101", "BUG-104"...]
  }
]
```

See It In Action

Three key workflows demonstrating intelligent bug triage.

01

Catch Duplicates

Submit a bug that semantically matches an existing one.

Input: "App crashes on login"

MATCHED: 92%

02

Auto-Enhancement

Submit a vague 3-word report and watch LLM expand it.

Input: "it doesn't work"

ENRICHED + STEPS

03

Visual Clustering

Identify a new outbreak of related issues on the map.

Action: View Cluster Map

FOUND: C1 (LOGIN)

Live Environment Active

Access at: https://demo.bug-triage-ai.com